

Volume 24

Sentinel Cyber Intelligence Engine (SCIE)

Version 1.0

Vision

The Cyber Intelligence Engine transforms raw observations into operational intelligence.

It answers questions like:

- What happened?
- Why did it happen?
- What changed?
- What is connected?
- What should I do next?
- What is likely to become a future problem?

Notice what it **does not** do:

It does **not** invent answers.

Everything must be supported by evidence already present in the platform.

Intelligence Pyramid

Raw Data

Everything flows upward.

Never downward.

Chapter 1

Five Levels of Intelligence

Level 1

Observation

Example

Server returned HTTP 301

Nothing more.

Level 2

Evidence

Server redirects HTTP to HTTPS.

Level 3

Knowledge

This website has always redirected correctly.

Now the platform understands history.

Level 4

Operational Intelligence

The redirect changed yesterday.

Now we understand context.

Level 5

Decision Support

No action required.

Now the system helps people.

Chapter 2

Intelligence Objects

The Intelligence Engine creates new object types.

Examples

Cyber Insight

Cyber Pattern

Cyber Trend

Cyber Recommendation

Cyber Prediction
Cyber Exception
Operational Risk
Trust Delta
Everything becomes searchable.

Chapter 3

Trend Engine

Instead of reporting today's state.
We report evolution.
Example
TLS Score

Another example
Missing Security Headers

Users see improvement.

Chapter 4

Pattern Engine

The Pattern Engine detects recurring behavior.
Example
Every deployment

Recommendation
Review deployment pipeline.

Chapter 5

Recommendation Engine

Recommendations are generated only when supported by evidence.

Each recommendation includes:

Reason

Evidence

Expected Benefit

Estimated Effort

Verification Plan

Confidence

Business Impact

Chapter 6

Trust Delta Engine

One of my favorite ideas.

Instead of showing only today's trust.

We also show change.

Example

Website Trust

Users immediately understand why.

Chapter 7

Risk Evolution

Instead of static risk.

We display movement.

Risk

Very useful for executives.

Chapter 8

Recommendation Prioritization

Recommendations are ranked using:

Risk Reduction

Implementation Cost

Verification Complexity

Business Impact

Dependencies

Confidence

Users focus on the highest-value actions first.

Chapter 9

Cyber Learning Engine

The platform learns about the customer's own environment.

Examples

Normal deployment schedule

Normal certificate renewal interval

Expected technologies

Expected cloud regions

Expected maintenance windows

Learning is scoped to the customer's authorized environment and remains explainable.

Chapter 10

Knowledge Evolution

Knowledge is never static.

Observation

The platform preserves context over time.

PS-ADR-0021

Intelligence Requires Evidence

Status

Accepted

Decision

The Cyber Intelligence Engine shall only generate insights that are supported by traceable evidence and documented reasoning.

Rationale

Trustworthy intelligence must be explainable, reproducible, and auditable.

NEW CORE ENGINE

Sentinel Dependency Intelligence Engine

This engine continuously discovers dependencies.

Example

Website

If one object changes.

Everything affected is immediately known.

NEW CORE ENGINE

Cyber Change Intelligence Engine

Instead of simply detecting changes.

It classifies them.

Examples

Expected Change

Approved Change

Unexpected Change

Risky Change

Unknown Change

Critical Change

This reduces alert fatigue.

NEW CORE ENGINE

Cyber Context Engine

This engine answers:

Why is this important?

Example

Finding

Missing CSP

Context

Public Website

Handles Payment Data

Business Critical

Compliance Impact

High

Priority automatically increases because of context, not because the rule changed.

NEW IDEA

Sentinel Cyber Timeline

Every object receives an interactive timeline.

Example

2028

This becomes the history of the object.

NEW IDEA

Cyber Story Generator

Instead of producing technical reports.

The platform can also generate an evidence-based narrative.

Example

During the last 90 days, your organization's web security posture improved significantly. Three high-risk issues were resolved, HTTP Strict Transport Security was enabled across all public websites, and certificate management has remained consistent. One new issue requires attention: a recently introduced API does not yet meet the organization's authentication policy.

Every statement is linked back to evidence.

This helps executives and auditors quickly understand the security journey without reading hundreds of technical pages.

NEW IDEA

Organizational Security Memory

This is an idea I have not seen implemented as a first-class concept.

Project Sentinel remembers organizational experience.

Examples:

- Which remediation approaches consistently succeed.
- Which findings frequently reappear.
- Which teams resolve issues fastest.
- Which deployment pipelines introduce regressions.
- Which policy exceptions remain open the longest.

This is not user profiling. It is operational learning that helps organizations improve their own processes.

Long-Term Vision: A Cyber Operating Environment

At this stage, I think we should refine our vision statement.

Instead of saying:

"Project Sentinel is a cybersecurity platform."

I propose:

Project Sentinel is an evidence-driven Cyber Security Operating Environment that helps organizations understand, protect, verify, and continuously improve their authorized digital environments.

Notice that this statement does **not** promise perfect security or claim to replace every specialized product. Instead, it clearly states what the system is designed to do—and it

aligns with every architectural decision we have made so far.

A strategic observation

We are no longer designing a collection of scanners.

We are designing four integrated layers:

1. **The Operational Layer** — missions, workflows, automation, and user experience.
2. **The Intelligence Layer** — evidence, knowledge, trust, relationships, and context.
3. **The Engineering Layer** — Python services, event bus, object model, APIs, plugins, and testing.
4. **The Security Layer** — web, endpoint, mobile, cloud, identity, network, compliance, and monitoring.

That separation of concerns will make Project Sentinel significantly easier to evolve over the next decade than many traditional security products, while staying fully implementable with Python, open standards, and modular engineering practices.

====[USER]====

Good and continue

====[ASSISTANT]====

Excellent. I think we are now ready to design what could become the **heart of Project Sentinel**.

Not the scanner.

Not the dashboard.

Not even the AI.

The **Mission Engine**.

Every security activity in Project Sentinel—from a home user checking a phone to a bank performing an enterprise assessment—will be represented as a **Mission**.

This is what makes the platform fundamentally different from products that expose dozens of disconnected tools.

PROJECT SENTINEL BLUEPRINT

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